

Bullet Farm: A Software-Change Transaction Processor for Heterogeneous Agents

Bullet Farm Maintainers

Abstract—Coding agents are now capable enough that their limiting failure is often not reasoning but the authority around it: two sessions edit one checkout, a review approves bytes that no longer exist, a timed-out push is retried after the server applied it, and a dashboard paints silence green. Bullet Farm is a software-change transaction processor *around* heterogeneous agents. Its binding chain—incarnation-fenced authority, one repository writer, private Attempt clones, exact-OID Candidates, independently bound Evidence, durable effect intent, read-back and adoption after ambiguity, and truthful UNKNOWN projection across five trust planes—is designed to remove seven concrete failure classes and to leave every integrated change with exact, replayable evidence of who observed which bytes under which gate. The upside, if it works, is verified autonomous software change whose cost, refusal rate, and unresolved unknowns are measured rather than inferred; none of it is claimed as measured here. Evidence is labelled **Designed**, **Component-proved**, **Transaction-proved**, or **Live/release-proved**. At snapshot 20260825-stage1-r3 nothing exceeds **Component-proved**: two bounded TLA+ safety models and the component lanes pass, while the connected **TRANSACTION PROOF**, every live provider/forged receipt, and the release are absent, and the release inventory reports 26 blocked of 26 evaluated gates. The biggest concerns are ranked with mitigations and retiring evidence: an unassembled connected product, absent live evidence, dependence on operator custody acts, a single-host Linux-only threat model, holdout-dependent verifier independence, a security score below its own floor, bounds the design cannot eliminate, and system size and cost. Comparison with a dated public corpus (Gas Town v1.2.1, Gas City v1.4.1, DeepSeek Harness dsh-v0.1.1-rc.2, Omnigent v0.10.0) makes no performance or superiority claim; a Stage-2 matched benchmark waits for the connected transaction proof.

Index Terms—coding agents, transaction processing, Git isolation, evidence binding, authority fencing, effect reconciliation, merge safety.

I. RESEARCH CONTRACT

Agent systems are capable enough that their limiting failure is often not model reasoning but the authority surrounding it. Classical transaction-processing work treats identity, concurrency, logging, recovery, and idempotence as one problem [1]–[3]; coding-agent stacks distribute those obligations among an editor, Git, CI, a provider process, and a forge without an end-to-end subject binding.

Bullet Farm’s thesis is narrower than “better agents”: it is a software-change transaction processor *around* heterogeneous agents, where a model may reason, search, and propose but cannot mint its own write, verification, or integration authority [4]. The research questions are:

- RQ1: Which identities and exact subjects must be bound so that a concurrent agent run can be recovered without inferring completion?

RQ2: Which bindings are implemented and component-proved in the current family, and which remain design or connected-product gaps?

RQ3: Do the reviewed pinned public contracts of adjacent systems document the same end-to-end bindings, after separating operational capability from authority and integrity?

RQ4: What observations would falsify the design, and what matched experiment would be required before a comparative performance claim?

A. Inclusion and Source Discipline

Deep studies were included when a system exposed a public architecture plus a release or commit suitable for reproducible inspection: Gas Town v1.2.1, Gas City v1.4.1, DeepSeek Harness dsh-v0.1.1-rc.2, and Omnigent v0.10.0 [5]–[8]. Issue reports were included only when they described a concrete authority, Git, evidence, or recovery failure mappable to a named invariant. The landscape appendix uses release/commit pins for open-source systems and official-document observation dates for proprietary ones; these are public-contract observations, not reverse engineering, and Codex statements rely only on official OpenAI documentation [9], [10].

Every comparison cell uses exactly one of five terms: Documented (a pinned primary source describes the property), Partial/configuration-dependent (it depends on configuration or covers part of the row), Not documented (the reviewed sources do not describe it, which does not assert the implementation lacks it), Unknown (the public material was insufficient to classify), and N/A (outside the product’s stated role). Absence of documentation is never a categorical “No.”

B. Evidence Vocabulary and Claim Limit

Table I is the paper’s claim lattice; the terms avoid the ambiguous word “complete” and are generated into both documents from `evidence.json`, `schema.bullet.paper-evidence.v1` [11].

The snapshot records doctor BLOCKED, fast lane PASS, demo PASS, and PASS for 2 bounded formal models. The release result is written as “26 blocked of 26 evaluated gates,” never as a bare fraction mistakable for success. UNKNOWN, timeout, skipped, unsupported, zero tests, a model assertion, HTTP 2xx, or process exit zero is not verified engineering truth [12], and no result from an active dirty tree enters the committed evidence manifest.

TABLE I
MATURITY LABELS USED THROUGHOUT THIS PAPER.

Label	Admitted meaning
Designed	Reviewed contract or invariant; no executable receipt required.
Component-proved	Exact committed component and mapped local receipt; no cross-process transaction inference.
Transaction-proved	One exact connected offline five-plane transaction with independent subject-bound receipts.
Live/release-proved	Admitted live provider/forged transaction, or signed release/install/recovery proof from the same subjects.

TABLE II
VALUE AND RISK AT A GLANCE, SNAPSHOT 20260825-STAGE1-R3.

Question	Answer at this snapshot
What it is	A transaction processor around coding agents: providers propose; a fenced kernel, one repository writer, an independent verifier, and an effect broker decide what is written, proved, and integrated.
Upside if it works	Removes the seven failure classes of Table III; every integrated change carries exact-subject Evidence naming who observed which bytes under which admitted gate; models and vendors can be swapped without granting any of them repository or merge authority; cost, refusal rate, and unresolved unknowns become measurable, with accepted, surviving change per human intervention and per dollar as the Stage-2 dependent variable, not a claim here; bounded multi-agent evolution is Designed and post-V1.
Proven today	Component-proved only: fast lane PASS, demo PASS, 2 bounded safety models PASS, doctor BLOCKED by design. TRANSACTION_PROOF ABSENT; live and release proof absent; 26 blocked of 26 release gates; audit 58 with 10 caps and 29 hard findings against a floor of 90/0/0.
Biggest concerns	(1) connected product unassembled; (2) no live provider or forged receipt; (3) closure depends on operator custody acts; (4) single host, Linux only, same-UID races open; (5) verifier independence rests on curated holdouts; (6) security score below floor; (7) split-brain, quota, and liveness bounded, not eliminated; (8) system size, cost, and forged adoption unmeasured.
Before launch	One signed connected transaction; operator-rated live admission and four provider receipts; Jeryu and GitHub read-back receipts; schema-3 lock and signed installer; audit at floor; five signed packages with SBOM and provenance; platform refusal receipts; then the Stage-2 matched benchmark.

C. Value and Risk at a Glance

Table II is the two-minute reading of this paper in the vocabulary of Table I; its numbers come from `evidence.json` or Table V, and Section VII-E expands the ranked concerns.

II. THREAT AND FAILURE MODEL

The V1 threat model is local-first and single-user: model output, repository content, concurrent agent incarnations, retrying supervisors, provider subprocesses, and network responses are untrusted, while the host’s admitted binaries and kernel primitives are trusted within explicitly tested boundaries. Remote multi-tenant runners and Byzantine forge operators are out of scope, and the same local UID remains a material limitation because it can race path-based tools that are not sealed or descriptor-relative.

The recurring failures are not peculiar to agents; they are ordinary distributed-systems failures expressed through Git:

- 1) *Authority aliasing*. Two living processes believe they own the same Attempt or writable checkout. A session name is not an incarnation fence.
- 2) *Repository aliasing*. Linked worktrees have per-worktree HEAD and index but share the object database, common

metadata, and most refs [13]; they improve ergonomics, not principals.

- 3) *Subject drift*. Tests or reviews name a branch, workspace, or PR rather than the exact base, head, tree, patch, environment, and gate set, so a rebase or merge silently invalidates the result.
- 4) *Ambiguous effects*. A timeout occurs after a server may have accepted a push or workflow creation; retrying before read-back duplicates the logical mutation.
- 5) *Inferred liveness*. `tmux`, a PID, an idle terminal, or a model’s “done” statement becomes a completion oracle, and silence is confused with death or success.
- 6) *Destructive recovery*. Cleanup removes a workspace before durable preservation and read-back establish whether the last mutation happened.
- 7) *Projection promotion*. A dashboard converts absent, stale, or contradictory ledger state into a green label and becomes accidental authority.

The public issue corpus supplies concrete instances: town-root checkout deleted operational files [14]; parallel rig-root writers clobbered edits [15]; a timeout after a server commit admitted duplicate workflows [16]; a non-atomic claim let two sessions execute the same task [17]; a session that died mid-task left its work hooked with no notification [18]; a patrol cleanup whose liveness check classified every worker as dead would have force-removed live worktrees [19]; and an aliveness check that trusted stale pid files never respawned dead refineries [20]. These reports are evidence for failure classes, not proof that the projects remain vulnerable at later heads.

A. The Transaction Subject

The minimum authority subject is not “agent a edits repository r ” but

$$A = (m, g, v, a, i, f, w, p, b),$$

where m is an immutable Mission revision, g its graph revision, v a Variant, a an Attempt, i the process incarnation, f its monotonically ordered fence, w a workspace generation, p a policy generation, and b a bounded budget reservation. A mutating request that omits one of these fields can alias a different logical run, and reclaim changes i and f , so every old request is stale even if its bearer still holds a valid-looking process handle.

The engineering subject is similarly richer than a commit hash:

$$C = H(\text{domain}, A, o_{\text{base}}, o_{\text{head}}, o_{\text{tree}}, h_{\text{patch}}, e, p),$$

where the OIDs are algorithm-tagged, e identifies admitted environment and tool subjects, and all variable-length inputs are framed. The formula is a design description, not a claim that every current constructor carries every term; it explains why equal content with different ancestry or authority need not be the same Candidate, why a moving branch is not a Candidate, and why a result for C says nothing about C' after a rebase, merge, policy change, or gate change.

An effect has its own identity rather than borrowing the Candidate identity:

$$X = (C, E, r, o_{expected}, o_{desired}, k, f),$$

where E is the independent Evidence root, r the protected destination, k the logical idempotency key, and f rechecked at the broker. The desired Git OID can be observed independently; the transport response cannot, so the ledger distinguishes command acceptance, local application, verification, effect intent, remote observation, and projection rather than collapsing them into one “done” bit that destroys the information needed for recovery.

B. Trust-Boundary Attacks

The model is intentionally hostile to ambient convenience. Repository files may request clean/smudge filters, external diff programs, merge drivers, credential helpers, hooks, submodules, alternate object stores, filesystem monitors, pagers, editors, or URL rewrites; provider output may contain terminal escapes, unbounded JSON, out-of-scope paths, or shell names disguised as gate requests; a runner may deliver a request twice, reorder a stale heartbeat after reclaim, or crash between local mutation and response; a forge may apply a write and lose the response; a projection may reconnect behind its watermark. Each is a typed protocol case, not an instruction to “try again.”

Conversely, Bullet Farm does not claim to defeat an administrator who can replace the running kernel, signing roots, or admitted binaries, nor to make a same-UID path race impossible where an external tool still resolves a mutable path. The design shrinks and names the trusted computing base rather than abolishing it, so release acceptance includes binary-digest admission, descriptor-relative/no-follow mutation, signed packages, provenance, and platform-specific containment instead of treating a source build as an installer.

III. FIVE-PLANE TRANSACTION ARCHITECTURE

Figure 1 shows the complete binding chain. The five planes are separated so that no plane can use its own assertion as evidence.

A. Control and Execution

The control plane owns Missions, immutable graph revisions, commands, leases, fences, policy snapshots, and durable outbox intent. A lease is insufficient without a monotonically changing incarnation fence checked by every mutation gateway [21], [22]; reclaiming expired work produces a new authority subject, and the old process cannot regain authority by waking up.

The execution plane supervises heterogeneous providers through admitted, bounded protocols: a provider receives read context and returns a typed patch proposal, never repository authority. Exactly one BulletGit daemon is the repository writer, and each Attempt uses a private clone rather than a linked worktree, eliminating shared index and common-ref control paths between Attempt principals. Hostile repository configuration, hooks, alternates, credential helpers, and

environment are input to be cleared or refused, not inherited ambient authority [23]. The committed policy names Landlock, seccomp, and cgroup v2 as the Linux production containment reference [24]–[26]; at the snapshot only the user and network namespace egress component exists [12], [27], so a bubblewrap-class filesystem and process sandbox for the provider [28] remains Designed.

B. Candidate and Verification

Applying a validated proposal produces an immutable Candidate identity over the provenance needed to reproduce it: base, head, and tree OIDs, canonical patch identity, Attempt/fence, and relevant environment and policy subjects. A content hash alone is insufficient because identical bytes can arise under different authority or ancestry; a branch name can move.

The verification plane reconstructs that exact Candidate in an independently clean environment with gates from an admitted catalog, not model-provided shell. Evidence binds the Candidate and gate IDs, executable/environment subjects, bounded output, exit status, and timing; any later subject change invalidates it. Independence is an authority relation, not a persona label.

C. Delivery, Audit, and Ambiguity

The delivery plane alone holds forge credentials. Before an effect it durably records logical effect identity, expected-old OID, intended-new OID, Evidence root, and idempotency key. Protected integration tests the exact merge result, following the merge-queue principle that what is tested is what becomes the branch [29], [30]; GitHub publication, when enabled, is a one-way mirror with expected-old-OID CAS, read-back, divergence quarantine, and independent backups, not a second writer.

If transport becomes ambiguous the command remains UNKNOWN and the broker reads the authoritative ref or remote object: an exact match is adopted with a receipt, a mismatch quarantined, and absence may admit a retry under the same logical effect identity. HTTP success is not verification and transport failure is not proof of non-application; this is the idempotent receiver discipline applied to Git effects [3].

The audit plane projects append-only events and exact watermarks. Portal state may be PENDING, APPLIED, VERIFIED, FAILED, or UNKNOWN; stale or contradictory streams stay visibly stale or unknown, and the plane cannot mint authority, repair evidence, or reinterpret missing data as success.

D. Worked Failure and Recovery Transaction

Consider Gas City issue 5511, where a timeout after a server-side commit led to additional workflows [16]. In the Bullet design: (1) a command and one effect identity are committed before dispatch; (2) a fenced Attempt in a private clone yields Candidate C ; (3) an independent verifier yields Evidence $E(C)$; (4) the broker records an intent to advance protected ref r from o to n ; (5) the network times out; and (6)

TABLE III
FAILURE CLASSES, UNSAFE SHORTCUTS, BINDING RULES, AND EXTERNALLY VISIBLE REFUSAL.

Failure class	Unsafe shortcut	Required binding	Honest observable outcome
Authority aliasing	PID, tmux name, or lease expiry treated as ownership	Attempt + incarnation + fence checked at every mutation	Stale principal is refused; current work remains unaffected
Repository aliasing	Parallel sessions write one checkout or common refs	Sole daemon writer and private generation per Attempt	Second writer has no repository capability
Subject drift	CI/reviewer names branch, PR, or working directory	Candidate exact base/head/tree/patch/environment; Evidence names Candidate	Rebase/merge/gate change invalidates Evidence
Ambiguous effect	Timeout classified as failure and retried as new work	Durable effect intent, one idempotency key, expected-old-OID CAS, read-back	UNKNOWN until exact applied/absent/divergent observation
False liveness	Idle terminal or missing heartbeat classified as completion/death	Lease/fence plus reconciled process and ledger observations	ALIVE, DEAD, UNKNOWN, or CONTRADICTORY
Destructive recovery	Force-remove before preserving unique bytes	Sealed preservation subject and durable cleanup/tombstone outcome	Cleanup refuses or returns UNKNOWN; recoverable bytes remain
Projection promotion	UI infers green from missing/stale state	Sequence/watermark-bound projection with no authority methods	Stale/absent/contradictory state remains visible

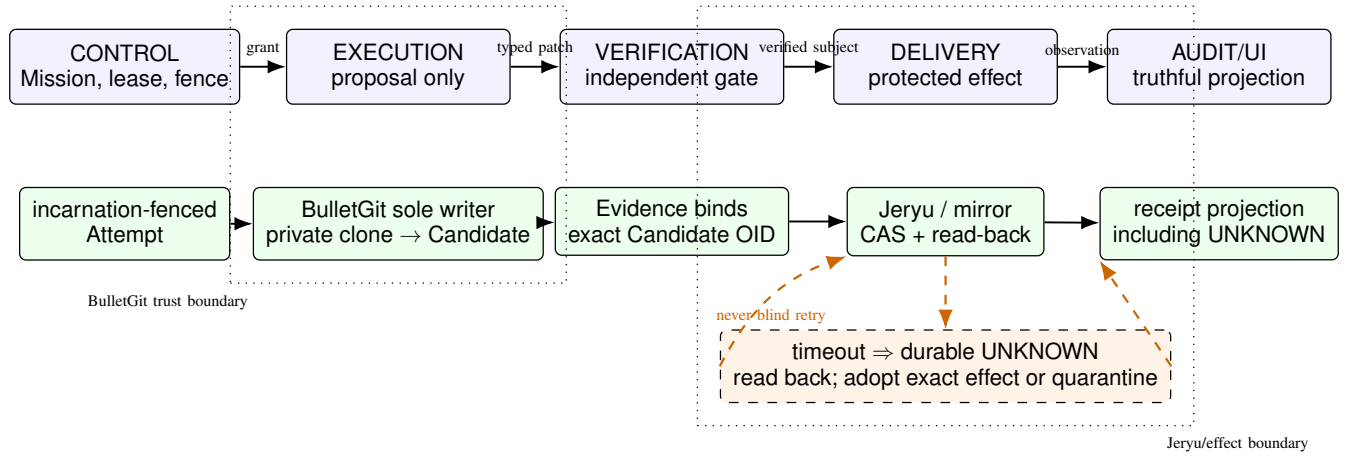


Fig. 1. Five trust planes and the cross-plane binding chain. A timeout after effect intent produces durable UNKNOWN; read-back either adopts the exact observed effect or quarantines divergence. It never authorizes a blind retry.

the command becomes UNKNOWN, not failed and not complete. Read-back then adopts the applied effect if $r = n$, may retry the same effect identity if $r = o$, and quarantines divergence otherwise. A new runner incarnation cannot create a second logical effect because ledger, fence, expected-old OID, and idempotency key are all checked at the mutation boundary.

The chain is the product: incarnation-fenced authority $\rightarrow$ sole repository writer $\rightarrow$ private Attempt clone $\rightarrow$ exact-OID Candidate $\rightarrow$ independently bound Evidence $\rightarrow$ durable effect intent $\rightarrow$ read-back/adoption $\rightarrow$ truthful projection. A sandbox, reviewer, or merge queue strengthens one link but does not imply the chain.

E. Cross-Plane Invariants

The architecture is constrained by invariants rather than agent roles; the registry distinguishes schema controls, gateway controls, and tests so that prose is never the only location of a safety rule. These ten are the compact paper view; the generated registry and closure plan remain the authoritative crosswalk [22], [31].

- 1) **One current incarnation.** At most one live incarnation/fence holds mutation authority for an Attempt; the successor's durable fence, not lease expiry, grants succession.
- 2) **One repository writer.** Provider, runner, verifier, portal, and effect processes lack direct workspace mutation capability.

Only BulletGit accepts a validated operation under the exact reservation.

- 3) **Private repository generations.** Attempt generations use no linked-worktree .git files or shared mutable control state, and prior-or-complete-next publication survives crashes.
- 4) **Scoped atomic patch.** Every operation is validated before any write. Traversal, .git, symlink, case-fold, normalization, duplicate, special-file, body-size, and operation-count ambiguity refuse the entire batch.
- 5) **Candidate identity is immutable.** Base, head, tree, patch, and authority/provenance subjects are length-framed and domain-separated. Content identity and provenance-bound Candidate identity are not conflated.
- 6) **Evidence is non-transferable.** The verifier reconstructs an exact Candidate with admitted gates. Any change to Candidate, gate, environment, policy, or verifier subject makes prior Evidence inapplicable.
- 7) **Attestor is not broker.** The principal that produces Evidence cannot use that act to mint protected-ref authority. The effect broker checks Evidence and current control authority independently.
- 8) **Effect identity survives retry.** Intent is durable before transport; expected-old OID, desired OID, idempotency key, and fence are read-back/adoption subjects, so retry never creates a second logical effect.
- 9) **Preservation precedes destruction.** Unique workspace bytes, journal identity, destination, and cleanup target are sealed and rechecked; post-delete ambiguity remains an outcome, not retrospective success.

- 10) **Projection is non-authoritative.** Commands return pending before settlement; sequence gaps, conflicting events, absent routes, and unavailable observations yield stale or unknown state; UI actions still pass through the authenticated command boundary.

These invariants keep the sovereign truths separate: the Kernel decides whether an operation is authorized, BulletGit what repository subject exists, the verifier whether Evidence for that subject satisfies admitted gates, and the broker whether an authorized, verified subject may be proposed to the forge, which remains sovereign over its protected ref. No single database row replaces those independent observations.

F. Evolutionary Control as Design

The intended product is a multi-agent system in which several Variants compete inside one Mission under the invariants above. Everything in this subsection is **Designed**: the normative model is a reviewed document, the committed policy fixes `evolutionary_authority=false`, and the closure plan gates the program on a connected transaction [22], [24], [32]. “Evolutionary” means bounded search over immutable, evidence-bearing software-change lineages—not autonomous credential use, self-reported fitness, unbounded debate, or online self-modification.

The genome is an immutable, content-addressed `TeamRecipe`: typed roles and contracts, communication edges and context policy, certified provider/profile choices, bounded budgets, and deterministic stopping, escalation, dissent, fusion, and fallback rules. It cannot contain or evolve credentials, authority claims, safety rules, risk classes, protected paths, evidence floors, hidden evaluators, integration policy, effect permission, or hard budget ceilings, and inherits no live session, credential, workspace, proof, or cache. A deterministic feasibility shield (authority, containment, certification, role conflicts, quota, budget, verifier capacity, task policy) runs before any objective, so infeasible recipes consume no budget and no score offsets a hard constraint. Eligible results enter a bounded quality-diversity archive in the MAP-Elites family [33], [34]: human-versioned, receipt-observable descriptors define cells, and each cell keeps a Pareto set over independently measured correctness, robustness, cost, latency, intervention, and observation survival with a content-ID tie-break. There is no global scalar reward.

Recipes advance only through completed, comparable rungs B0–B6 in the successive-halving discipline of staged budgets [35], [36]: deterministic checks, a small then a larger matched visible block against the incumbent under the same total compute, a sealed confirmation under independent oracle custody, no-effect shadow routing, a bounded canary with automatic rollback, and an eligible routing arm with an incumbent traffic reserve. Raw holdout failures never enter author context; every task class keeps a strong single-agent incumbent; agent count, transcript volume, confidence, and activity are never fitness dimensions; a sparse island model copies only immutable recipe identity, re-evaluated at the destination [37]; and a monitored bound breach removes a recipe and falls back to the incumbent. V1 records routing and

outcome provenance and does not update weights, prompts, policies, or privileges online; contextual routing learners [38] are post-V1 and require holdouts with unbiased offline evaluation [39], rollback, drift detection, and an explainable signed decision record. None of these mechanisms has a receipt; they are absent by design rather than incomplete.

G. BulletGit as the Repository Trust Boundary

BulletGit does not replace Git’s object model or pack protocol; it wraps ordinary Git with a capability and evidence boundary. A bare mirror can amortize network/object transfer, but an Attempt clone dissociates before use so no surviving alternates file makes mirror garbage collection part of workspace correctness, and the clone loses its origin and ambient authentication. Git child processes receive a scrubbed environment and explicit configuration that disables inherited global/system config, paging, hooks, credentials, external attributes, signing, editors, and prompts; repository-scoped commands first inspect local configuration with includes disabled, and command-bearing or truth-redirecting values fail closed [23].

This matters because a repository is active input: a clean filter executes during checkout, `textconv` alters what a diff appears to show, a merge driver creates bytes outside the intended patch, URL rewriting or a remote helper redirects network authority, a credential helper makes a private clone push-capable, alternates make object availability depend on another repository, and hooks turn a read-only Git verb into execution. A provider sandbox does not neutralize those semantics, especially when the provider itself runs Git.

Patch application validates the complete batch before mutation: paths are canonical normalized relative bytes with no parent traversal, `.git` component, NUL, backslash ambiguity, alternate-data-stream syntax, trailing dot/space, symlink traversal, or folded collision; operation and body counts are bounded; checkpoints use a temporary index rather than the live workspace index; and a Candidate is minted only after Git reports the expected tree/head relationship and the journal durably identifies the generation.

Preservation is part of the protocol rather than cleanup hygiene: a sealed receipt binds source workspace, recoverable artifact, destination, journal, Candidate/Attempt subjects, and exact cleanup target; the daemon re-verifies receipt and filesystem object immediately before deletion; a synced tombstone records the settled outcome; and failure after deletion but before confirmation is typed UNKNOWN, never certainty reconstructed from absence.

The current production constructor is intentionally fail-closed: positive online final-authority settlement is unavailable, so the public mutation path refuses before repository I/O. That refusal is useful component evidence, not evidence that the designed positive path works; enabling it requires an immutable signed shared-wire subject, locally verified grants, current Kernel reservation/fence read-back, short operation permits, and durable settlement of the same transaction.

H. Jeryu Source Authority and Blocked Topology Proposals

The frozen V1 topology distributes source Jeryu-only from immutable signed tags. It assumes neither `git.neverhuman.org` nor any GitHub namespace or public mirror, the family manifest admits no authenticated Jeryu URL, the only permitted local Jeryu family is `/home/ubuntu/jain-split/jeryu-split`, and `/home/ubuntu/jeryu-split` must never be recreated.

The named `git.neverhuman.org` deployment and the `github.com/neverhuman/jeryu` and `github.com/neverhuman/jankurai` front doors are unratified, operator-blocked proposals, not current topology or authority. A later operator choice may treat GitHub as a public index or backup and host selected open-source families on independent Jeryu Git; neither name is source authority today. A ratified public mirror remains a separately configured effect adapter with expected-old-OID CAS, read-back, divergence quarantine, and independently proved backups—a mirror is not a backup.

Jeryu’s local hub provides branded onboarding, the family manifest, and pinned component mapping while component repositories remain independent. Jankurai is an existing monorepo with no assumed public front door; before any external-history or mirror decision the rule is preservation first (complete ref and object bundles, ancestry map, reconciliation RFC, rehearsal, explicit approval), because a force-push discards the evidence needed to decide which history survives. The local-family rule is frozen; endpoint and mirror proposals remain **Designed** and operator-blocked until dated ratification precedes TLS, authentication, protection, push/check/read-back, backup/restore, deployment, and mirror receipts.

IV. IMPLEMENTED EVIDENCE AND CONNECTED GAPS

Table IV separates current component receipts from the connected claims they do not establish. Repository subjects come from the shared evidence snapshot (Hub 3039878371b6, Kernel ca380bc4ffd4, BulletGit fb715b25c3b2, Portal 95108e37d643); a later source commit invalidates mapped receipts until replay, and publication-only descendants are admitted only when preflight proves that evidence-critical inputs did not change.

The table is intentionally asymmetric: many component tests coexist with zero connected-product proof. `just demo` is a deterministic demonstration, not a live transaction; `doctor` is **BLOCKED** because the checked family lock is intentionally unsupported and stale; the release inventory is a truthful negative receipt, not a score.

At the Kernel subject the live-admission boundary is mechanized on both sides and closed: the Kernel policy loader mirrors the Hub’s `v1alpha2` rule (live admission only at policy generation ≥ 2 with an active `provider-runner` key; commit 0d848f6), the nightly lane’s real-binary mode refuses to start without an operator-ratified policy path (b4735da), and under the committed generation-1 policy every provider is refused at `POLICY_LIVE_ADMISSION_DISABLED` before any key read, probe, or spawn, with default nightly halves targeting a marker script so that any spawn fails the lane.

Ratification is an operator act recorded in the family ledger; no provider has a live receipt [12], [22].

A. Evidence Lifecycle

An admissible receipt answers five questions: *who observed what exact subject, using which admitted mechanism, under which boundary, with what outcome?* Test counts answer none of them, so the snapshot stores repository commit and tree, clean-state fact, tool versions, complete command, exit code, semantic outcome, and a sentence stating what the result does not prove, and hashes the family manifest, family lock, release truth, invariant registry, formal locks, policy, and competitor snapshot, all outside the PDF hash loop.

Evidence moves monotonically only within one immutable subject: a new commit, tree, policy generation, executable, gate, environment, or dependency lock does not make the old receipt false, only inapplicable. A dirty-tree replay is useful diagnosis but cannot be committed as frozen evidence, because the tested bytes are not named by the commit; a generated release report is a projection of receipts and blockers and cannot satisfy one of its own rows.

Component receipts are still valuable: Kernel tests can falsify a lease or command implementation, hostile Git fixtures a path or configuration rule, browser tests a projection, and bounded model checks a protocol invariant within its bounds. The discipline is to preserve those results without an invalid conjunction: “Kernel passed” $\wedge$ “BulletGit passed” $\wedge$ “Portal passed” does not imply that a signed command traveled through all of them with one Candidate and one observed effect.

B. Formal Models and Their Boundary

Exactly two bounded TLA+ models are pinned and checked with TLC [40], [41]; the lock fixes the TLC release by digest, the Java major, single-worker execution, and the module, configuration, and state counts, and the lane is **PASS** for 2 models. *LeaseFence* models two runners, one lease, a monotonically increasing fence, a database clock, scope revisions behind a mutation barrier, freeze, and restore epochs (`MaxFence=3`, `MaxTime=4`, `MaxScope=3`, `MaxOps=2`, `MaxEpoch=2`, `MaxFreeze=3`), checking exactly `TypeOK`, `BarrierHasNoInFlightApply`, and `RestoredAuthorityNeedsFreshAcquire` over 529,851 generated and 128,913 distinct states at depth 24. *Stale-fence*, *stale-epoch*, *expired*, *frozen*, and *barrier* applies are refused by the authorization guard and counted rather than asserted, and fence monotonicity holds by construction, not by check. *EffectCheck* models one effect and one dependent check against a remote that may be untouched, desired, or third-party-written (`MaxDispatches=2`, `MaxEpoch=2`), checking exactly `TypeOK`, `AtMostOneLogicalEffect`, `VerifiedEffectWasReadBack`, `CheckRequiresDurableProof`, and `ThirdPartyStateIsNeverAdopted` over 1,585 generated and 378 distinct states at depth 18. `AtMostOneLogicalEffect` is enforced by assignment

TABLE IV
IMPLEMENTED NOW VERSUS NOT YET PROVED AT SNAPSHOT 20260825-STAGE1-R3.

Boundary	Maturity	Implemented evidence	Not established
Wire and admission	Component-proved	Canonical signed DTOs, bounded IPC, executable-digest and launch-grant components	Immutable shared-wire publication and connected issuer/consumer settlement
Control authority	Component-proved	DB-clock leases, fences, atomic command/outbox, authenticated ingress, quarantined signed lease-transport, proposal-to-checkpoint binding	Durable Mutation reservation/settlement and one connected Kernel-Runner-BulletGit issuer path
Provider boundary	Component-proved	Four offline message subsets, policy-first refusal, bounded process and Linux egress components	Any live provider receipt; policy remains live-disabled in the frozen evidence subject
BulletGit	Component-proved	Fail-closed production gateway, private-clone/preservation primitives, cleanup UNKNOWN, strict local manifests	Positive online sole-writer transaction; public clone remains AUTHORITY_CONTRACT_UNAVAILABLE
Verifier	Component-proved	One admitted fixture gate, exact-subject Evidence, bounded transport failure handling	Production reconstruction source, executable digest for every gate, and multi-gate aggregation
Effects	Component-proved	Command/effect ambiguity model and local bare-forge components	Authenticated Jeryu/GitHub expected-old-OID push and read-back receipts
Portal/audit	Component-proved	Generated contracts, authenticated command reconciliation, watermark-bound projections, explicit absent-surface UNKNOWN	Packaged runtime and complete Candidate/Evidence/Effect ledger subjects
Security audit	Component-proved	Deterministic score 58 (raw 58), 10 caps, 29 hard and 15 soft findings	V1 floor ≥ 90 , zero caps/hard findings, and a portable pinned hosted artifact
Connected product	Designed	Normative five-plane transaction and release contracts	Transaction-proved or Live/release-proved evidence; 26 blocked of 26 evaluated release gates

TABLE V
STAGE-1 COMMAND FAMILIES, RECORDED OUTCOMES, AND INTERPRETATION. FULL COMMANDS AND BOUNDARIES ARE IN `EVIDENCE.JSON`.

Command family	Snapshot outcome	What it establishes	What it cannot establish
bullet-family doctor -json	BLOCKED	Fail-closed install/lock diagnosis	A supported installed family
bullet-family check release -json	BLOCKED	26 named blockers among 26 evaluated gates	Any positive gate or percentage score
Hub just fast	PASS	Mapped Hub component checks at the subject	Sibling, transaction, live, or release truth
Hub just demo	PASS	Deterministic multi-process demonstration semantics	TRANSACTION_PROOF; provider/forge/release evidence
Hub contract lane	PASS	Wire/schema generation and drift checks	Immutable publication or consumer settlement
Invariant registry	PASS	Machine crosswalk and enforced-entry validation	Implementation of rows whose lifecycle is planned
Local audit lane	PASS component; release blocked	Score 58/raw 58, 10 caps, 29 hard and 15 soft findings	V1 quality floor or portable hosted execution
Repository required lanes	PASS	Mapped committed components for Kernel, BulletGit, and Portal	Cross-repository composition
Formal model lane	PASS	2 bounded models match pinned sources/tools/counts	Liveness or implementation correctness
Paper build/preflight	deterministic publication check	Shared values, citations, layout, metadata, embedded fonts, page size/count, reproducible PDF bytes	Product correctness or release authority

rather than by counting physical writes, so the dispatch guard carries the real protection.

The effect model also produced a refutation rather than a decorative pass. A sixth candidate, `NoDispatchAfterStop`—“after policy expiry or freeze, nothing is dispatching”—is *false* as a state predicate, and TLC exhibits the counterexample at depth 4: `Init` $\rightarrow$ `PersistEffectIntent` $\rightarrow$ `DispatchEffect` $\rightarrow$ `ExpirePolicy`, symmetrically `Freeze`. The trace is the real system property: a stop cannot recall a request already on the wire, so the honest guarantee is that no *new* dispatch leaves after a stop. The module therefore keeps that predicate defined but unchecked as a documented boundary and checks the action property

```
NoNewDispatchAfterStop ==
  [][(~policyLive \ / frozen) =>
    UNCHANGED «effectDispatches,
    checkDispatches»]_vars
```

which is the regression guard on the two dispatch actions. State counts are unchanged at 1,585 generated and 378 distinct at depth 18, and the both the module and configuration digests

are re-pinned in the model lock [40]. An invariant list that asserted the state predicate would have been false; the model caught that, which is the argument for bounded models over a prose claim of safety.

Both configurations set `CHECK_DEADLOCK FALSE`; no liveness or fairness property beyond that action property is checked. Wound-wait [42], multi-resource scope expansion, two Attempts on one Variant, Git, provider protocols, the forge, OS containment, and multi-tenant attackers are not modelled, and the JSON traces beside the models are executable fixtures replayed by Rust tests, not model-check results. Model success remains **Component-proved**, never **Transaction-proved**.

C. Open Connected Product Boundary

The authoritative closure plan, not this paper, controls V1 [22]. The remaining predecessor chain is an immutable `bullet-wire` tag that consumers can honestly resolve, a Kernel reservation that repeats lease checks inside one write, production `BulletGit` verification of that permit, connected Candidate/Evidence settlement, authenticated protected inte-

gration, provider conformance, packages, installer, backup, signing, provenance, and live/release receipts. The workplan is only a backlog that loses to that authority [43]; the absent end-to-end TRANSACTION_PROOF is not hidden behind component totals.

V. PINNED SYSTEMS AND CAPABILITY MATRICES

The comparison asks two questions: operational capability (what work a system helps users perform) and authority/integrity (what exact subject and principal the public contract binds). Combining them would make a rich orchestrator look unsafe merely because it is featureful, or a strict component look complete merely because it refuses unsafe work.

A. Deep Case Studies

Gas Town v1.2.1. Gas Town provides a legible multi-agent town: role-specific agents, Beads-backed work, tmux sessions, hook worktrees, patrol and recovery machinery, and a Refinery merge queue [5], [44]. Its strength is operations: many agents and providers supervised through durable work records. Its public contract uses worktrees and orchestration roles rather than a separate repository-writer principal, and linked worktrees, which share the common repository and most refs, are not private repository principals [13].

Gas City v1.4.1. Gas City generalizes the factory with Agents, Beads, Formulas, Rigs, Packs, Events, multiple runtimes, retry/fan-out/drain controls, and release artifacts [6], [45]. The richer runtime and workflow vocabulary should not be minimized. The issue audit nevertheless shows why session durability and exactly-once effects are distinct: a shared rig root, a non-atomic task claim, and retry after ambiguous creation all cross different trust boundaries [15]–[17].

DeepSeek Harness rc.2. DeepSeek Harness is a Cordis plugin kernel: model, tools, sandbox, storage, session, loop, scheduling, and UI are composed as plugins, with model-visible context reconstructed from an append-only session log [46]. It is not categorically single-agent: its experimental Agent Teams seam adds a roster, task board, mailbox, and continuable child agents, while its documentation warns that checkout scopes are advisory rather than locks [47]. Its primary contribution is composable agent execution, not protected Git integration.

Omnigent v0.10.0. Omnigent wraps several vendor harnesses behind a uniform session/runtime layer and documents Linux bubblewrap and macOS seatbelt sandboxes plus an L7 proxy that keeps credentials outside the sandbox and can transform network requests [48], [49]. Contextual policy verdicts are ALLOW, DENY, and ASK; transformation belongs to the proxy, not the verdict [50]. Polly adds parallel worktrees and cross-vendor implementation/review, a meaningful operational feature [51], [52]; the wrapped harness still performs repository actions, so approval attribution and Candidate binding remain separate questions.

B. What the Architecture Should Borrow

The comparison is not a feature-stripping exercise. Gas Town demonstrates the operator value of named roles, inspectable work records, patrol, recovery, and a dedicated

landing function; Gas City makes runtimes and workflow structures explicit and adds release-operational depth. A transaction processor that cannot explain where work is, recover a session, limit dispatch, or expose a usable queue is safer only by being unusable, so Bullet’s control plane should compose these concepts while never letting a role name substitute for an authority subject.

DeepSeek Harness shows that provider, tools, storage, sandbox, loop, and UI can be composable plugins, and its append-only model-visible session log matters: if context cannot be reconstructed from durable events, recovery silently changes the agent’s inputs. Bullet can consume a plugin kernel behind the provider boundary as long as plugin composition cannot acquire repository or effect authority. Experimental Teams shows subagent coordination and a task board as protocol objects; the difference is enforcement, since advisory checkout scopes coordinate while a transaction gateway must refuse overlapping mutation.

Omnigent demonstrates defense in depth at a different layer: native OS sandboxing, a credential-isolating L7 proxy, contextual approval, shared sessions, and cross-vendor review remain desirable alongside sole-writer and Candidate bindings. Bullet need not reimplement that UX; it can wrap or accept proposals from those systems provided the proposal is bounded untrusted data and all repository/effect credentials stay across the gateway, while an L7 proxy can transform network traffic without becoming the authority that says a Candidate passed its gates.

The interoperability rule follows: protocols such as ACP and MCP may carry sessions, context, and tool calls but do not mint Bullet capabilities. A provider adapter may translate a typed proposal, a review adapter return commentary, and an MCP tool expose read-only repository facts; none may cross from “the model requested” to “the repository is authorized” without the same signed control subject, current fence, admitted operation, and durable settlement used by native components.

C. Classification Procedure

For each cell the review fixed the release/commit and source set, looked for an explicit contract, and assigned the least affirmative supported term. A feature shown only in an optional pack, plugin, hosted configuration, or experimental seam is **Partial/configuration-dependent** unless the row is about that seam; **Not documented** records a search result, not an implementation claim; **Unknown** marks a public description too ambiguous even to call the property undocumented; and **N/A** is reserved for products whose public role excludes the property, such as a plugin kernel that does not claim to operate a protected forge.

The Bullet design and Bullet Stage-1 rows are separate so that a design document is not compared against released competitors as if it were running code. The Stage-1 row is **Partial/configuration-dependent** whenever an exact component exists but the positive connected path does not: a fail-closed unavailable constructor proves that unauthorized writes are refused, not that a repository writer operates. The terms are not ordered numerical values and must not be summed.

D. Why Familiar Mechanisms Do Not Automatically Compose

One objection is that every part of the chain already exists: worktrees isolate files, containers isolate processes, CI verifies commits, merge queues protect branches, and idempotency keys prevent duplicate requests [53]. The premises are mostly correct; the conclusion fails unless the subjects and principals at their boundaries are the same.

A worktree is not a writer identity. It gives a per-worktree HEAD and index, yet common repository metadata and most refs remain shared [13], and an agent that can invoke arbitrary Git can still name refs, helpers, remotes, hooks, config, objects, and sibling worktrees outside the intended edit surface. A private clone with authentication removed is not magical isolation, but it makes the repository principal a separable object that the sole writer can validate and destroy.

A sandbox is not a repository transaction. A container, bubblewrap [28], seatbelt profile, or remote VM restricts filesystem and network access; it does not decide whether the bytes written are an admitted patch for the current fence, whether tests name the resulting tree, or whether a protected push applied after timeout. Sandboxing complements rather than replaces Candidate and effect identity.

A durable session is not durable engineering state. Transcripts, mailboxes, task boards, tmux sessions, and event logs make reasoning resumable and may prove what an agent saw or said; they do not prove that the repository contains a particular tree, that a gate ran against it, or that the forge accepted it. Session history serves cognition; repository truth comes from Git and verification truth from Evidence.

A reviewer is not automatically an independent attestor. A second model catches semantic mistakes and cross-vendor review reduces correlated failure, but independence still fails if both reviewers share a mutable checkout, the author can rewrite the test oracle, the result names only a branch, or the reviewer credential can execute the protected effect. The authority requirement is reconstructible exact-subject verification and attestor/broker separation; model diversity is additional quality control.

A merge queue is not effect reconciliation. Testing the exact merge result is a central integrity property [29], [30], but the queue still needs a caller identity, current authority, an Evidence subject, expected-old ref, and a rule for lost responses; an item reported failed after its merge applied is duplicated by an upstream retry unless logical effect identity survives transport ambiguity.

An idempotency key is not useful unless the receiver binds it. A token held only in process memory disappears on restart; a receiver that records it after rather than atomically with its effect leaves a crash window; a token reused for a different desired OID launders a conflict. The effect record must bind the logical command, Candidate, Evidence, protected ref, expected and desired OIDs, current fence, and observed outcome, durably before and after transport [3].

Bullet Farm’s novelty claim is therefore compositional and falsifiable, not that it invented locks, hashes, queues, or idempotence. If a smaller public contract demonstrates the same bindings under the fault corpus, Bullet’s extra machinery is

unnecessary; until then, comparing one isolated mechanism with the entire chain is a category error.

The result is not that Bullet Farm is universally superior. It has less live product evidence than every mature system in this table. The defensible result is that the reviewed pinned public contracts do not document the same end-to-end bindings. Whether those bindings justify their cost is precisely the Stage-2 benchmark question.

VI. ISSUE-TO-INVARIANT AUDIT

Table VIII records state and affected/reported versions as observed on 2026-08-25. “Pinned corroboration” distinguishes issue testimony from code/release material at the selected subject. An issue being closed does not erase the failure class; an open issue does not prove current pinned code is affected.

The audit corrects four tempting misreadings: issue 742 is about a settings path and supports no respawn-count claim; DeepSeek Teams makes categorical “single-agent” language false; Omnigent policy returns ALLOW/DENY/ASK, and proxy transformations are a different layer; and linked worktrees do not share one index, since each has its own index and HEAD while common metadata and most refs are shared [13].

VII. EVALUATION, FALSIFICATION, AND ARTIFACT

A. Stage-1 Evaluation Protocol

Stage 1 is architecture and component assurance. From one dated evidence snapshot the artifact records repository commit/tree/clean facts, toolchain versions, full commands, doctor and release-gate outcomes, component and formal receipts, external pins, and hashes of evidence-critical inputs; both PDFs consume generated macros from that JSON, and the strict preflight of Appendix B deliberately fails while any canonical family checkout is dirty.

The reproducible command surface from the Hub is:

```
just paper just paper-check
cargo run -locked -quiet -bin
bullet-family - doctor -json
cargo run -locked -quiet -bin
bullet-family - check release -json
just fast just demo
```

Repository-mapped required/contract/formal commands and their observed outcomes are enumerated in evidence.json; summary macros do not replace them. The Hub subject excludes publication-only descendants to avoid a self-hash loop; preflight admits such a descendant only when every changed path is in the paper/tooling set and all hashed evidence inputs are identical.

B. Fault-Oriented Acceptance Corpus

A happy-path patch is a weak discriminator, since nearly every reviewed system can edit a file and run a test; the connected acceptance corpus must inject faults at every state boundary and retain the resulting ledger, Git, Evidence, effect, and observation subjects. At minimum it includes:

- 1) reclaim an Attempt while its old provider and runner remain alive, then deliver stale mutation, heartbeat, verification, and effect messages;

TABLE VI
OPERATIONAL CAPABILITIES IN PINNED PUBLIC CONTRACTS. THESE CELLS DO NOT IMPLY AUTHORITY SAFETY.

System	Orchestration	Provider composition	Isolation	Durable sessions	Review	Merge queues
Bullet Farm Stage 1 [4], [22]	Documented	Partial/configuration-dependent	Partial/configuration-dependent	Documented	Partial/configuration-dependent	Partial/configuration-dependent
Gas Town v1.2.1 [5], [44]	Documented	Documented	Partial/configuration-dependent	Documented	Partial/configuration-dependent	Documented
Gas City v1.4.1 [6], [45]	Documented	Documented	Partial/configuration-dependent	Documented	Partial/configuration-dependent	Documented
DeepSeek Harness rc.2 [46], [47]	Documented	Documented	Partial/configuration-dependent	Documented	Documented	Not documented
Omnigent v0.10.0 [48], [51]	Documented	Documented	Documented	Documented	Documented	Partial/configuration-dependent

TABLE VII
AUTHORITY AND INTEGRITY PROPERTIES. NOT DOCUMENTED MEANS ONLY THAT THE REVIEWED PINNED PUBLIC CONTRACT DOES NOT STATE THE BINDING.

System	Writer identity	Incarnation fence	Exact verification subject	Effect idempotency/read-back	Protected integration	Truthful uncertainty
Bullet Farm design [4]	Documented	Documented	Documented	Documented	Documented	Documented
Bullet Farm Stage 1 [12], [22]	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent	Documented
Gas Town v1.2.1 [5], [44]	Partial/configuration-dependent	Not documented	Not documented	Not documented	Partial/configuration-dependent	Partial/configuration-dependent
Gas City v1.4.1 [6], [45]	Partial/configuration-dependent	Not documented	Not documented	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent
DeepSeek Harness rc.2 [46], [47]	Not documented	Not documented	Not documented	N/A	N/A	Partial/configuration-dependent
Omnigent v0.10.0 [48], [50]	Partial/configuration-dependent	Not documented	Not documented	Not documented	Partial/configuration-dependent	Partial/configuration-dependent

TABLE VIII
PRIMARY ISSUE EVIDENCE MAPPED TO BULLET INVARIANTS AND CURRENT MATURITY.

Project / ID / state / version	Pinned-code or release corroboration	Failure class	Mapped Bullet invariant	Current maturity	Residual risk
Gas Town #4094; open; reported 1.1.0 [14]	v1.2.1 retains town/hook worktree architecture; issue is not treated as a pin-wide verdict	Repository/config aliasing	Sole writer; hostile Git/config refusal; private Attempt clone	Component-proved	Connected BulletGit authority and same-UID path races remain open.
Gas Town #742; closed; reported before v1.2.1 [54]	v1.2.1 was inspected; no “687 respawns” figure is present or claimed	Configuration placement	Sealed admitted HOME/config and exact executable subject	Component-proved	Live provider configuration and every real-provider receipt remain absent.
Gas Town #3584; closed; version not stated [18]	v1.2.1 README keeps the Problems-view row “Zombie: Dead tmux session” as the liveness signal [44]; issue records a mid-task death that left work hooked without notification	Inferred liveness (silent death)	Lease/fence heartbeat with runner self-fence; hooked work never completes by silence	Component-proved	DB-clock leases and self-fence exist; the four-valued reconciled observation is designed.
Gas Town #4737; open; reported 1.1.0 [19]	v1.2.1 retains patrol and cleanup molecules; issue records a liveness check that classified every dog dead before forced worktree removal	Inferred liveness (false death) and destructive recovery	Destructive eligibility only from exact incarnation identity; preservation before deletion	Component-proved	Preservation and tombstone are local components; no connected cleanup settlement.
Gas Town #4091; open; version not stated [20]	v1.2.1 retains daemon aliveness checks; issue records stale pid files trusted as liveness	Inferred liveness (stale identity)	Incarnation fence and ledger observation, never PID or session name	Component-proved	Fence is a component; the reconciled ALIVE/DEAD/UNKNOWN/CONTRADICTION observation is designed.
Gas City #1181; open; version not specified [15]	v1.4.1 documents rigs/packs; issue records parallel agents resolving to one rig path	Shared writable identity	Private per-Attempt clone; one writer principal	Component-proved	No connected positive BulletGit transaction.
Gas City #5511; open; affected v1.4.1 [16]	Exact selected release is the reported affected version; issue records post-commit timeout and duplicate workflows	Ambiguous effect retry	Durable effect identity; expected-old-OID CAS; read-back/adoption	Component-proved	No authenticated live forge receipt or transaction proof.
Gas City #1052; closed via #2985; affected 0.15.2 [17]	v1.4.1 is later than the reported fix; retained as historical class evidence only	Non-atomic claim	Atomic lease/fence/Attempt command transition	Component-proved	Signed online transport is not connected end to end.
Omnigent #1694; open; affected 0.2.0 [52]	v0.10.0/Polly materials document parallel worktrees and cross-vendor review	Session/config ambiguity	Durable Attempt state; incarnation fence; preservation before cleanup	Component-proved	Provider and reviewer identities are not yet live-bound in Bullet.
Omnigent #725; open; version not stated [55]	v0.10.0 scope was inspected; issue concerns external/native edits in a non-Git workspace	Incomplete change observation	Candidate manifest and honest UNKNOWN for unrepresented change state	Component-proved	Non-Git workspaces are outside Bullet V1; Git Candidate path is not connected.
Omnigent PR #4318; merged; included in v0.10.0 [8], [56]	Release notes identify the approval-attribution roll-back as a breaking change	Credential/approval attribution	Separate requester, approver, broker, and effect identities	Designed	Operation-specific components exist, but no live owner-credential effect proves separation.

- 2) crash before and after patch journal sync, Candidate publication, Evidence settlement, effect-intent sync, remote application, response receipt, read-back, preservation, deletion, and tombstone sync;
- 3) rebase or merge between verification and integration, change a gate executable or policy generation, and attempt to reuse stale Evidence;
- 4) inject hostile Git config, hooks, filters, alternates, attributes, credentials, helpers, submodules, symlinks, path normalization/case collisions, large frames, invalid Unicode, terminal escapes, and protocol contamination;
- 5) drop, duplicate, reorder, truncate, and delay every JSON-RPC and command response, including a server-applied forge write whose client response is lost;

- 6) replace an admitted pathname after validation, race cleanup targets, exhaust descriptor-relative traversal limits, and exercise each supported or explicitly refused operating system; and
- 7) disconnect/reconnect the projection across sequence gaps, conflicting events, missing routes, and restore epochs, verifying that no display becomes authoritative.

The acceptance criterion is not merely “no crash”: every injected boundary must end in exact prior state, exact complete next state, durable failed state with preserved subject, or truthful UNKNOWN/quarantine requiring reconciliation, and no outcome may contain two current fences, two logical effects, Evidence for the wrong Candidate, an unpreserved unique

workspace, or a projection greener than its source.

Formal and implementation traces should cross-check without being confused: the two TLA+ models generate small safety traces for lease/fence and effect ambiguity, Rust tests replay them against component state machines, and the connected corpus must repeat equivalent faults across real processes, filesystem boundaries, and a controlled forge. Agreement is stronger than any layer alone, yet remains one offline Transaction-proved subject until live admission repeats the same binding.

C. Stage-2 Matched Benchmark

Stage 2 is prohibited until a signed connected TRANSACTION_PROOF exists. It will freeze identical repositories and issue tasks; pin every system, model, tool, policy, budget, and retry rule; run enough trials to expose tail failures; and retain exact final trees, tests, human adjudication, wall-clock time, token/cost measures, collisions, duplicated effects, ambiguous outcomes, recovery actions, and data loss, with operational success and authority violations as separate dependent variables. Systems unable to emit a native receipt will be wrapped by an external observer without pretending the wrapper is a product feature. Only that matched corpus can support “faster,” “cheaper,” “safer,” or “more successful”; this paper makes none of those claims.

Matched evaluation must also account for refusal: a system that declines every mutation has no integrity failures and no utility, and one that always continues looks fast by discarding ambiguity. Results therefore report task utility, safe refusal rate, recovery latency, operator interventions, and unresolved unknowns together, with private-clone storage, verifier reconstruction time, merge-queue dwell, and read-back traffic as first-class costs.

D. Falsification Conditions

The architecture would be weakened or rejected by any of the following:

- a stale or reclaimed incarnation mutates a repository or remote effect;
- two Attempts share a repository principal or common mutable control path;
- Evidence remains valid after any Candidate subject changes;
- timeout/restart can create two logical effects or erase an applied one;
- cleanup destroys the only recoverable bytes before preservation settles;
- Portal or prose can promote UNKNOWN to verified;
- the same guarantees are achieved by a substantially smaller public contract under the matched fault corpus; or
- bookkeeping/latency cost dominates benefit under realistic workloads.

The paper itself is falsified by a source row that does not support its cell, an unrecorded dirty-tree result, a snapshot mismatch, a non-reproducible PDF, or a claim above its maturity, even if the Rust components remain correct.

E. Biggest Concerns, Limitations, and Artifact

The eight concerns below are ranked by how much of the stated upside each withholds. Each names the mitigation

present at the snapshot and the evidence that would retire it; this paper retires none of them.

- 1) *The connected product is unassembled.* No positive row exceeds **Component-proved**; TRANSACTION_PROOF is ABSENT and 26 of 26 release gates are blocked [12]. Mitigation: fail-closed gates that prose cannot promote. Retired by one signed connected five-plane transaction, then live and release receipts from the same subjects.
- 2) *No live provider or forge receipt exists at the subject.* The committed policy refuses every provider before spawn, one provider has a deep fake-process proof, and no authenticated Jeryu or GitHub read-back exists [22], [24]. Mitigation: the boundary is mechanized on both sides. Retired by sealed conformance receipts for all four providers and exact-subject forge integration/read-back receipts.
- 3) *Closure depends on operator custody acts.* Live-admission ratification, Jeryu authentication, release signing keys, and the schema-3 lock cannot be produced by engineering; each has a typed refusal and a runbook, and only the recorded act and its receipt retire this.
- 4) *Single host, Linux only, same UID.* Multi-tenant runners, hostile hosts, model-output semantics, non-Git workspaces, and non-Linux mutation are out of scope; some path-selected Git/helper subjects remain raceable by the same user. Mitigation: sealed, descriptor-relative subjects where implemented; other platforms refuse. Retired by the fault corpus plus per-platform containment and refusal receipts.
- 5) *Verifier independence rests on curated holdouts and one fixture gate.* One admitted gate, no per-gate executable digest, no multi-gate aggregation; oracle-modifying diffs and any later evolutionary search are only as strong as the sealed holdouts a team maintains [32]. Mitigation: attestor/broker separation, and holdout failures never reach authors. Retired by a production gate catalog with digests and receipted holdout custody.
- 6) *Security is below its own floor.* Audit 58 (raw 58), 10 caps, 29 hard findings, no portable hosted artifact. Mitigation: the lane fails closed and ratchets upward only. Retired by a pinned receipt at ≥ 90 with zero caps and hard findings.
- 7) *Bounded, not eliminated.* Split-brain is bounded by lease, fence, and self-fence, not removed; quota is vendor-capped, so UNKNOWN is safe but never known; the formal models check bounded safety, not liveness [40]. Mitigation: typed UNKNOWN and CONTRADICTORY outcomes with read-back. Retired only as far as the fault corpus and benchmark show the bounds hold under load.
- 8) *Size, cost, and adoption.* Five planes and four repositories are a large integration surface; private clones cost storage, and fencing, exact subjects, and read-back add latency; only Jeryu and a GitHub adapter are specified. Mitigation: invariant tiering and the falsification condition that a smaller contract refutes the design. Retired by Stage-2 measurements reporting cost, refusal rate, and

unknowns together.

Beyond these, public documents can omit private controls, issues are selected counterexamples rather than incidence statistics, proprietary systems are black boxes, and open-source heads move after their pins. The artifact is this source, the executive brief, bibliography, architecture vector, evidence JSON and schema, generated macros, deterministic build/preflight scripts, and the non-authoritative workplan; the frozen V1 topology and the operator-blocked endpoint and mirror proposals of Section III-H are not live receipts [43].

VIII. CONCLUSION

Powerful agents increase the need for narrow, inspectable authority. Bullet Farm’s proposed contribution is neither a stronger persona nor another agent loop but the composition of familiar transaction mechanisms over exact Git subjects, bound at every boundary to the same principal and the same bytes. The evidence boundary is equally important: the family has meaningful component receipts and bounded models, while the connected transaction, every live provider/forgo proof, and the release remain absent. The reviewed pinned public contracts do not document the same end-to-end bindings, but that is a contract comparison, not measured superiority. Stage 1 publishes the architecture, its ranked concerns, and its falsifiable gaps; Stage 2 begins only after the exact connected transaction exists.

APPENDIX A

DATED LANDSCAPE BEYOND THE DEEP STUDIES

Table IX is an observation index, not a capability score: open-source entries name the exact inspected commit, and proprietary entries name official documents observed 2026-08-25, black-box limited.

Pins are observation subjects, not endorsements or claims about later versions; for proprietary systems only the cited official surface is classified, and undocumented properties remain Unknown.

APPENDIX B

EVIDENCE MANIFEST AND PUBLICATION PREFLIGHT

The machine source is `docs/paper/evidence.json`, schema `bullet.paper-evidence.v1: snapshot ID`, UTC time and source epoch; four repository commit/tree/clean subjects; tool versions; full commands with exit code, semantic outcome, and limitation; shared result values; the four-term maturity vocabulary; dated external pins; and hashes of evidence-critical inputs. The build validates that object and emits `evidence.generated.tex`, which both documents consume, so gate counts, repository abbreviations, audit results, and absent transaction status are never manually duplicated. PDFs do not appear in their own input manifest, since embedding a PDF hash in source that changes the PDF is circular; successful preflight reports final hashes as output receipts.

Strict preflight rejects a dirty capture or checkout, false maturity, repository/commit/tree mismatch, a Hub head that is not a paper/tooling-only descendant of the snapshot subject

with unchanged evidence hashes, a Kernel, BulletGit, or Portal mismatch, a stale input hash, an undefined or multiple citation/reference, a duplicate or unused bibliography entry, and an external reference without URL/DOI and access date; it builds twice under one `SOURCE_DATE_EPOCH`, UTC, deterministic metadata, and locale with repeating PDF hashes, and requires no material overfull box, US Letter, full metadata, embedded fonts, 12–14 paper pages, and exactly four brief pages. Every rendered page is then inspected, because no log parser proves that a table, URL, or vector arrow is legible. This is publication assurance, not a product gate: a byte-identical paper can faithfully report a broken product, and a correct component can coexist with a failed paper check.

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TABLE IX
LANDSCAPE SNAPSHOT OBSERVED 2026-08-25. FULL 40-HEX PINS ARE IN THE BIBLIOGRAPHY AND EVIDENCE MANIFEST.

System	Pinned subject or official observation	Public-contract scope and limitation
Codex	Official CLI/cloud docs [9], [10]	Local CLI and hosted delegated tasks are documented; public docs do not establish Bullet's connected authority chain. Proprietary/hosted behavior is black-box limited.
Claude Code	Official docs observed 2026-08-25 [57]	Agentic terminal/editor workflow; proprietary behavior and service controls are black-box limited.
Cursor	Official agent docs observed 2026-08-25 [58]	IDE/agent workflow; proprietary internals are black-box limited.
GitHub Copilot	Official coding-agent docs observed 2026-08-25 [59]	Hosted issue-to-PR workflow integrated with GitHub; service internals are black-box limited.
Jules	Official Google docs observed 2026-08-25 [60]	Hosted asynchronous coding tasks; service internals are black-box limited.
Devin	Official docs observed 2026-08-25 [61]	Hosted coding-agent sessions; service internals are black-box limited.
Kiro	Official docs observed 2026-08-25 [62]	IDE/spec-driven agent workflow; service internals are black-box limited.
Gemini CLI	commit 812f7a2bcf20 [63]	Open-source terminal agent; Apache-2.0.
OpenHands / Agent Canvas	commit 150e76046db0 [64], [65]	Open agent platform and control surface; MIT. Agent Canvas documents isolated worktree operation and ACP composition.
SWE-agent	commit 3ea751c087f3 [66]	Research issue-solving harness; MIT.
mini-SWE-agent	commit 25941c89cfeb [67]	Minimal research harness; MIT.
Aider	commit 5dc9490bb35f [68]	Open terminal pair-programmer; Apache-2.0.
Cline	commit 4f5f2384076e [69]	Open editor agent; Apache-2.0.
Continue	commit 5522c6f44ca0 [70]	Open IDE/agent framework; Apache-2.0.
Goose	commit f1d32e8dbbdf [71]	Open extensible agent; Apache-2.0.
Roo Code	commit b867ec914575 [72]	Open editor agent; Apache-2.0.
ACP	commit 5acaa3acd11 [73]	Agent-client interoperability protocol; Apache-2.0; not a repository authority system.
MCP	commit 57ac4a2ec742 [74]	Model-tool/context interoperability protocol; MIT; not a repository authority system.

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APPENDIX C RED-TEAM SELF-REVIEW

Ten attacks a hostile reviewer will run, each answered or conceded.

- 1) “*Your issues do not say what you say.*” Conceded for an earlier draft that attributed a respawn storm and a “687 respawns” figure to issue 742, which concerns a settings path and has no source for the figure. The liveness class now cites issues 3584, 4737, and 4091 with state, date, and pinned corroboration (Table VIII); residual: trackers move.
- 2) “*Component receipts are not a product.*” Conceded; nothing here exceeds Component-proved, 26 of 26 release gates are blocked, the connected transaction is absent, and Section VII-E ranks that first.
- 3) “*Two TLA+ models prove the system.*” Conceded: bounded safety only, deadlock checking off, one invariant true by assignment, no wound-wait, Git, forge, or provider state. The models are not decorative, though: TLC refuted NoDispatchAfterStop as a state predicate at depth 4 and forced the weaker, true action property.
- 4) “*Fencing tokens, idempotency keys, sandboxes, and merge queues with new names.*” Largely conceded and cited; the claim is compositional (same subject and principal at every boundary) and its worth is the Stage-2 question.
- 5) “*Providers propose is unenforceable.*” Partly conceded: only a validated proposal crosses the writer boundary, but the provider sandbox is Designed, only the network-namespace egress component exists, and no transaction has composed admission, proposal, and apply.
- 6) “*An evolutionary paper with no evolutionary code.*” Conceded: Section III-F is design, the policy pins evolutionary_authority=false, and the program is post-V1 and gated on the connected transaction.
- 7) “*A single-user loopback threat model is trivial.*” Narrow by design: the adversaries are the model, a concurrent incarnation, a retrying supervisor, hostile repository content, and a promoting projection—the corpus failures. Multi-tenant and hostile-host threats are out of scope.
- 8) “*An agent can flip live admission.*” Rejected: the Hub validator enforces the schema rule and the Kernel loader mirrors it, live admission needs generation ≥ 2 and a registered runner key, real-mode nightly refuses without an operator-ratified policy path, and an environment token was rejected as admission.
- 9) “*A security paper with 29 hard findings.*” Conceded; the figures are generated, dated, and block release by the family’s own rule, and no security quality is claimed.
- 10) “*The comparison is a cartoon of featureful systems.*” Rejected: the operational and authority matrices are separate, Not documented is never converted into “No,” each deep study credits what the system contributes, and no superiority language precedes the benchmark.